K-FIELD: CUMULATIVE HYSTERESIS ON THE CLOSED EARTH-CMB ROUTE

Forward and Reverse Circulation Through an Asymmetric Scalar Cost Field

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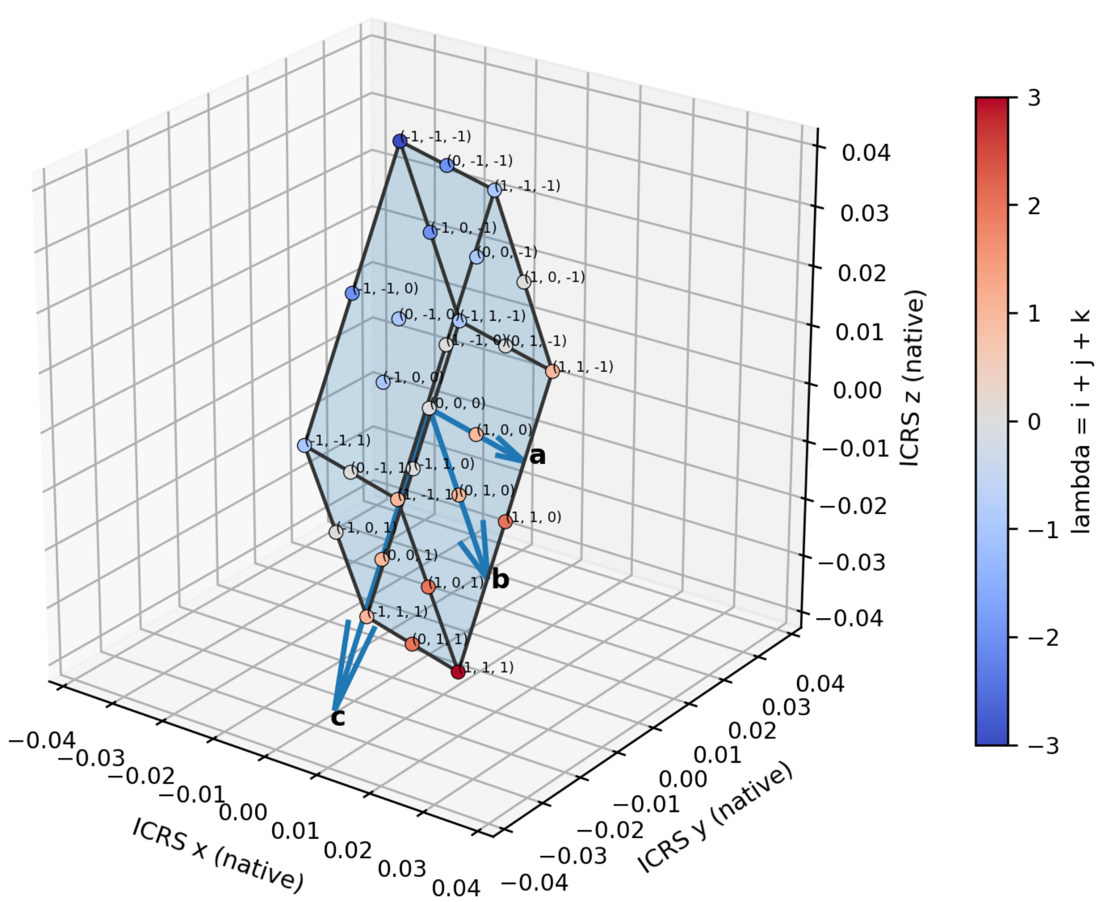
Krytronx

Authoritative Cell3: eight vertices, all 27 half-step nodes, and six face planes



Title figure. Dimensionally correct authoritative Cell3 in native ICRS Cartesian coordinates. The eight boundary vertices, all 27 centered half-step nodes, three primitive edges, and six exterior face planes are generated directly from the authoritative basis. Node labels are the integer addresses (i,j,k) obtained by doubling the stored half-step coefficients. Cell3 native scale is kept separate from the unit route sphere.

Archival foundational education reference - Unix epoch 1784658249

Abstract

This paper develops the complete geometry and cumulative-cost mathematics of a closed wire coil wound along the smoothed Earth-CMB route in the authoritative ICRS quadratic phase metric. The route consists of a high-cost meridian from the CMB antapex toward the apex, a low-cost return meridian, and two finite-radius sinusoidal-curvature joining tracks that replace the endpoint cusps. The wire therefore forms one continuous differentiable loop while preserving 99.941 percent of the original meridian cost asymmetry. The local scalar density is q(n)=sqrt(n^T G_phase n). Its completed closed-loop integral is unchanged by reversing circulation. Nevertheless, forward and reverse circulation encounter the spatial density in different orders. The cumulative integrals C_F(s) and C_R(s), compared after equal traveled distance s, form a closed hysteresis loop rather than retracing one curve. The maximum separation is 2.748220082602 normalized cost units at exactly half the loop. The signed enclosed area is 7.596421328302 normalized cost-distance units and the absolute enclosed area is 7.617373149318. The central distinction is therefore exact: equality of the final closed-loop scalar integral does not imply equality of the intermediate accumulated state. The local field is scalar and memoryless, but the accumulated traversal state is ordered and history dependent. This geometric cumulative hysteresis provides a rigorous route by which phase delay, threshold response, nonlinear impedance, switching, or any other stateful transport law can distinguish circulation direction without requiring the bare scalar line integral itself to be direction odd.

Central result

Forward and reverse circulation finish with the same total cost, C_F(L)=C_R(L)=11.276251390281, but they do not follow the same cumulative trajectory. At s=L/2, C_F=7.012235736442 and C_R=4.264015653840. Their separation is 2.748220082602, or 24.3718 percent of the completed-loop cost.

Contents

1. Scope, Coordinate Domains, and Definitions

The construction contains two geometries that share one orientation frame but do not share one length scale. The closed route lies on a normalized direction sphere of radius R=1. Its coordinates are unit ICRS direction vectors. Cell3 is a separate native lattice cell expressed in the same ICRS Cartesian axes. Its native basis magnitudes are approximately 0.0208, 0.0365, and 0.0493. The route sphere is not a magnified Cell3, and Cell3 is not assigned the route radius. Keeping these domains separate prevents a sky-direction coordinate from being mistaken for a physical lattice displacement.

The route coordinate s is arc length measured in units of R. When R=1, s is numerically equal to angular distance in radians. The scalar density q is dimensionless. Consequently, the normalized cost C is numerical. At a physical radius R_phys, physical distance is R_phys s, physical cumulative cost is R_phys C, and the area between cumulative traces scales as R_phys squared.

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PRIMARY DEFINITIONS
ICRS route state: n(s) in R^3, with ||n(s)|| = 1
Local scalar density: q(n) = sqrt(n^T G_phase n)
Forward cumulative cost: C_F(s) = integral_0^s q_F(u) du
Closed-loop length: L = 6.223020710975859 R
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2. Authoritative Quadratic Phase Metric

The K-Field phase density used throughout this paper is generated by the symmetric positive-definite matrix G_phase. Positive definiteness makes q(n) real and strictly positive for every nonzero direction. The matrix eigenvalues are all positive and span a factor of 6.15697, so the unit direction sphere carries a strongly anisotropic but smooth scalar cost landscape.

row	G_x	G_y	G_z
0	2.9764106378	1.2431507795	-2.0803600354
1	1.2431507795	2.5922286635	-2.3178839937
2	-2.0803600354	-2.3178839937	6.7735709133

metric property	value
eigenvalues, ascending	1.429328753487, 2.112545026447, 8.800336434665
trace	12.342210214600
determinant	26.572803745831
spectral condition number	6.156971524705

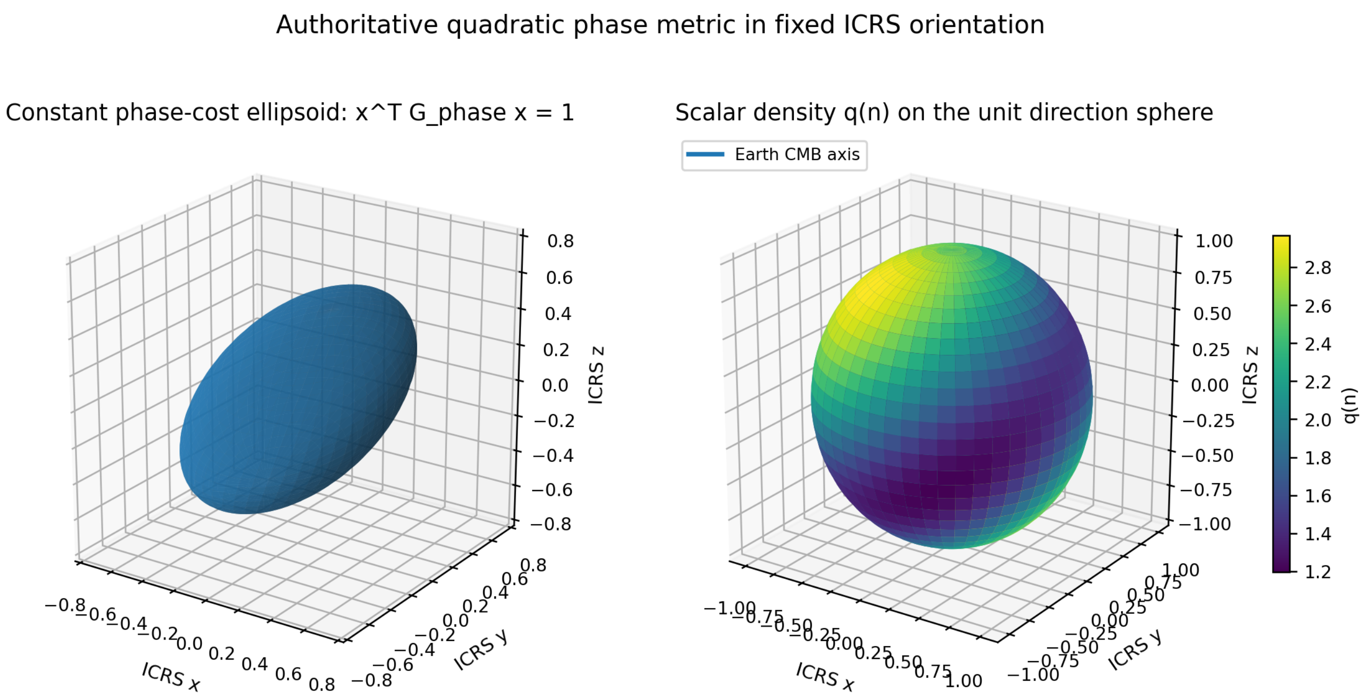

Figure 1. The left panel is the constant-cost ellipsoid x^T G_phase x=1. The right panel shows q(n) evaluated over the unit ICRS direction sphere, with the Earth-CMB diameter drawn through the field. Both surfaces are generated directly from the supplied 3 x 3 metric.

3. Cell3 Reference Geometry in Native ICRS Units

The authoritative Cell3 basis is included because it fixes the native K-Field lattice orientation and its primitive face systems. It is a reference geometry, not a replacement scale for the route sphere. Let B=[a b c]. Every centered half-step node is x(i,j,k)=0.5(i a+j b+k c), where each of i,j,k belongs to {-1,0,1}. This produces exactly 27 nodes. The eight boundary vertices are the cases in which all three indices are nonzero.

basis vector	ICRS x (native)	ICRS y (native)	ICRS z (native)	magnitude
a	0.020381438516946	-0.003416414496411	-0.002332965916729	0.020797058782433
b	0.002198745968323	0.012902170258693	-0.034066501536056	0.036494205130727
c	-0.000630379485347	-0.024257722017128	-0.042924354571094	0.049308565899859

face family	area (native^2)	included angle (deg)	unit normal ICRS	support planes n dot x = +/- h
A_ab	0.000754721789673	83.933483012313	(0.194092200808, 0.913177168821, 0.358379234791)	+/- 0.018828573500193
A_ac	0.001011255757688	80.448130018338	(0.089052594102, 0.866576769291, -0.491033948326)	+/- 0.014052166903391
A_bc	0.001385784431733	50.363231813756	(-0.995965190309, 0.083602162507, -0.032619290564)	+/- 0.010254361619054

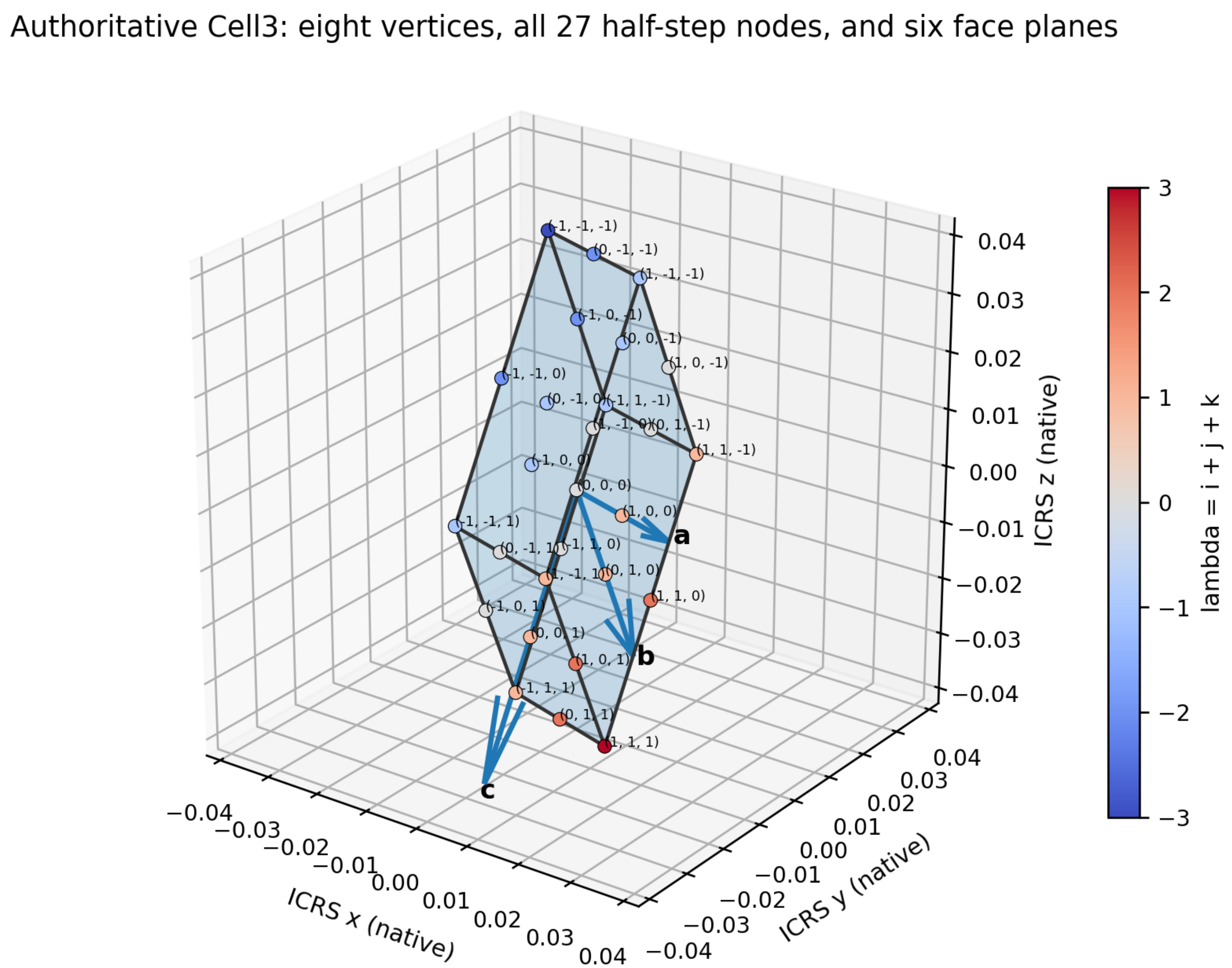

Figure 2. Authoritative Cell3 in native ICRS units. The six translucent quadrilateral surfaces are the primitive opposite-face planes. Every one of the 27 centered half-step nodes is plotted and labeled by the integer address (i,j,k).

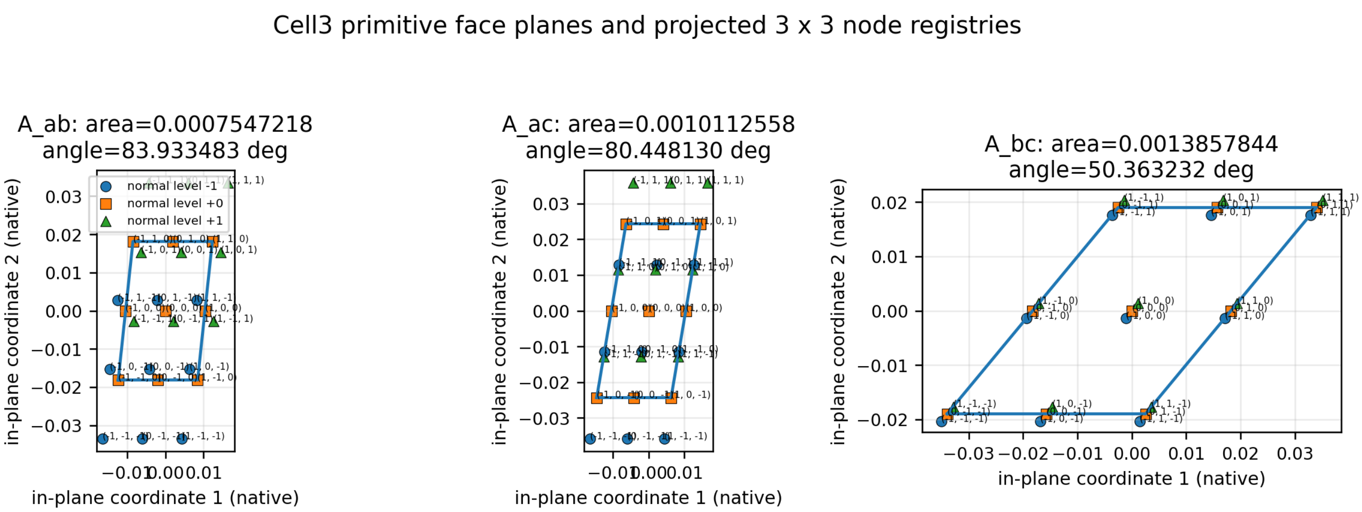

Figure 3. Orthographic projections into the three primitive face families. The three marker forms represent the three levels along the missing basis coordinate, so each panel shows how the full 27-node registry collapses onto a 3 x 3 in-plane address pattern.

4. Closed Earth-CMB Route Geometry

The common diameter of the route is the Earth-CMB axis c_hat=(-0.970770855835935, 0.207348009110986, -0.120874929481992). Its apex is at RA 167.943299879828 deg, Dec -6.942599922130 deg; the antapex is the exact antipode. The two meridian planes share this diameter but differ in plane orientation by 89.769450000754 deg.

Before smoothing, the high-cost route is the optimized antapex-to-apex meridian and the return is the minimum-cost apex-to-antapex meridian. Each is a semicircle on the unit sphere. Their plane normals and in-plane basis directions are supplied directly in the source packet.

route component	direction	plane normal ICRS	unsmoothed cost	mean q
high-cost meridian	antapex_to_apex	(0.131553362649, 0.880925815222, 0.454602486631)	7.053342663751	2.245148700514
minimum-cost return	apex_to_antapex	(0.200211790745, 0.421865187990, -0.884276541591)	4.310950793504	1.372218256424

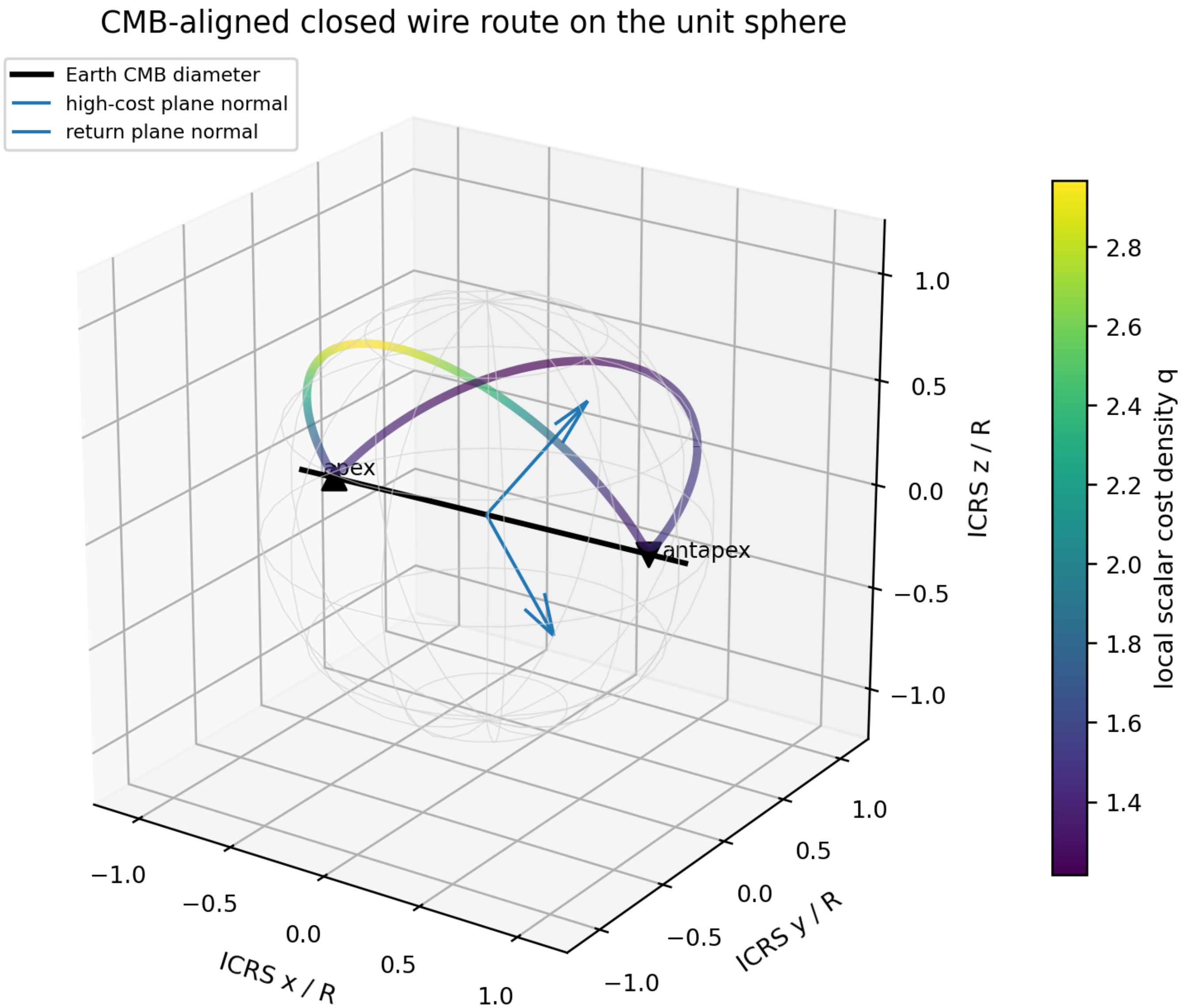

Figure 4. The continuous wire route on the normalized unit sphere. Color is the local scalar cost density q. The black line is the common Earth-CMB diameter. The route-plane normals are shown from the origin. This figure uses the authoritative ICRS orientation without RA/Dec reversal or display-frame remapping.

4.1 Finite-radius endpoint joining curves

The two optimized meridians meet at the apex and antapex with a tangent rotation of 89.769450000754 degrees. A direct splice would therefore create a cusp and an undefined instantaneous curvature. The closed roller-coaster construction trims 6 degrees from each end of both meridians and inserts a symmetric sinusoidal-curvature bridge in the tangent plane at each endpoint. The bridge is mapped back to the unit sphere by the exponential map.

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SMOOTH JOINING LAW
Bridge geodesic curvature profile: k_g(u) proportional to sin^2(pi u / ell), 0 <= u <= ell
k_g(0)=k_g(ell)=0 and dk_g/ds(0)=dk_g/ds(ell)=0
bridge arc length ell = 0.179364667368454 R
peak geodesic curvature = 17.470246433754 / R
minimum distance from endpoint axis = 1.614456340326 deg
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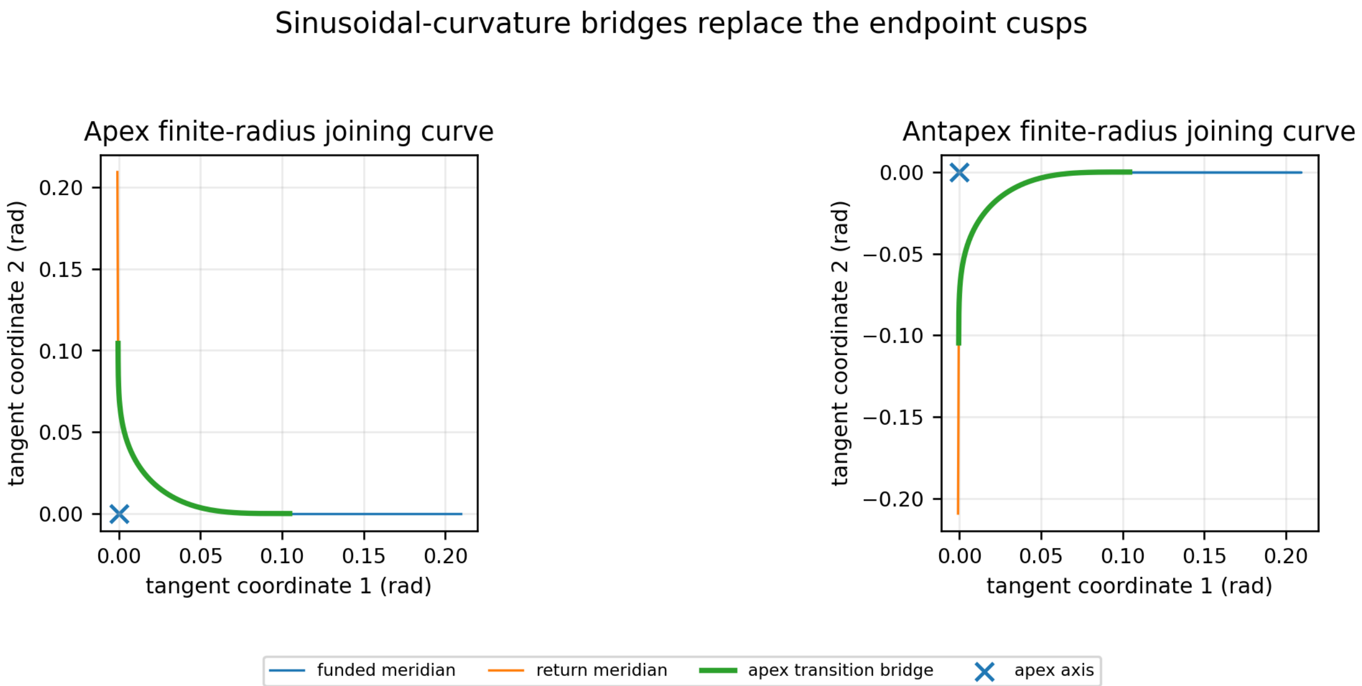

Figure 5. Local tangent-plane views of the apex and antapex joining tracks. The bridge rotates continuously between the two meridian tangents while avoiding the endpoint axis by a finite distance. The curve has zero added geodesic curvature and zero curvature slope at both interfaces.

5. Wire Coil and Scalar Traversal Cost

A wire wound on the exact centerline inherits the same four ordered segments: high-cost meridian, apex bridge, return meridian, and antapex bridge. For one turn, the normalized closed-wire length is 6.223020710976 R. The scalar cost is integrated using periodic trapezoidal quadrature over 14,400 equal intervals. The duplicate endpoint is removed before periodic integration and restored only after the cumulative arrays are closed.

segment	length / R	integrated cost	mean q	min q	max q	samples
funded meridian	2.932166355831	6.745302483472	2.300450133076	1.377373991790	2.966349502993	6785
apex transition bridge	0.179343999657	0.266933252970	1.488386862570	1.438192141845	1.574408143887	415
return meridian	2.932166355831	4.004524463371	1.365722123988	1.215958608102	1.520023276661	6785
antapex transition bridge	0.179343999657	0.259491190469	1.446890840874	1.395538189710	1.485534366837	415

The trimmed high-cost meridian contributes 6.745302483472; the return contributes 4.004524463371. Their difference is 2.740778020101, and the high-to-low ratio is 1.684420346329. The smoothed route retains 99.94115173 percent of the original unsmoothed asymmetry.

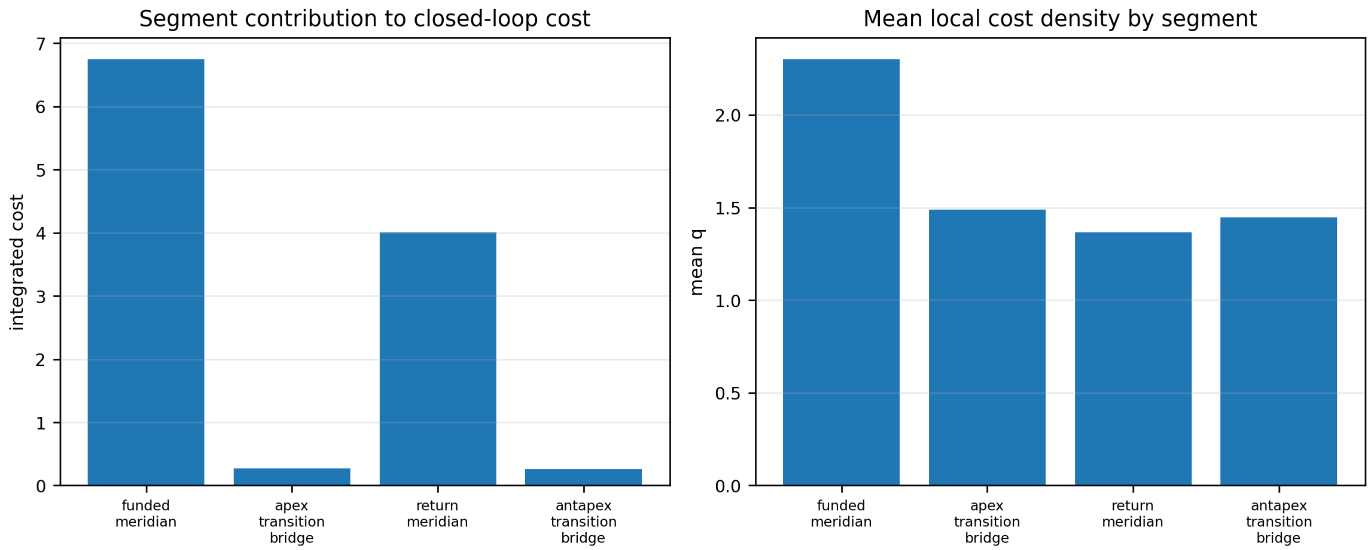

Figure 6. Integrated cost and mean local density for the four geometric segments. The high-cost meridian dominates the total cost; the two bridges are short and contribute only small corrections.

6. Forward and Reverse Cumulative Circulation

Choose the start point s=0 on the closed wire and orient increasing s in the forward direction. The local forward sequence is q_F(s)=q(n(s)). Reverse circulation starts from the same physical point and moves in decreasing forward coordinate. Its equal-distance local sequence is q_R(s)=q_F((-s) mod L). The two sequences contain the same values over a complete lap, but generally in different order.

```
FORWARD AND REVERSE CUMULATIVE FUNCTIONS
C_F(s) = integral_0^s q_F(u) du
C_R(s) = integral_0^s q_F((-u) mod L) du
C_R(s) = C_loop - C_F(L-s)
C_F(L) = C_R(L) = C_loop
```

The completed-loop equality follows by a change of integration variable. It is an equality of the final scalar integral only. It does not impose equality at intermediate traveled distance.

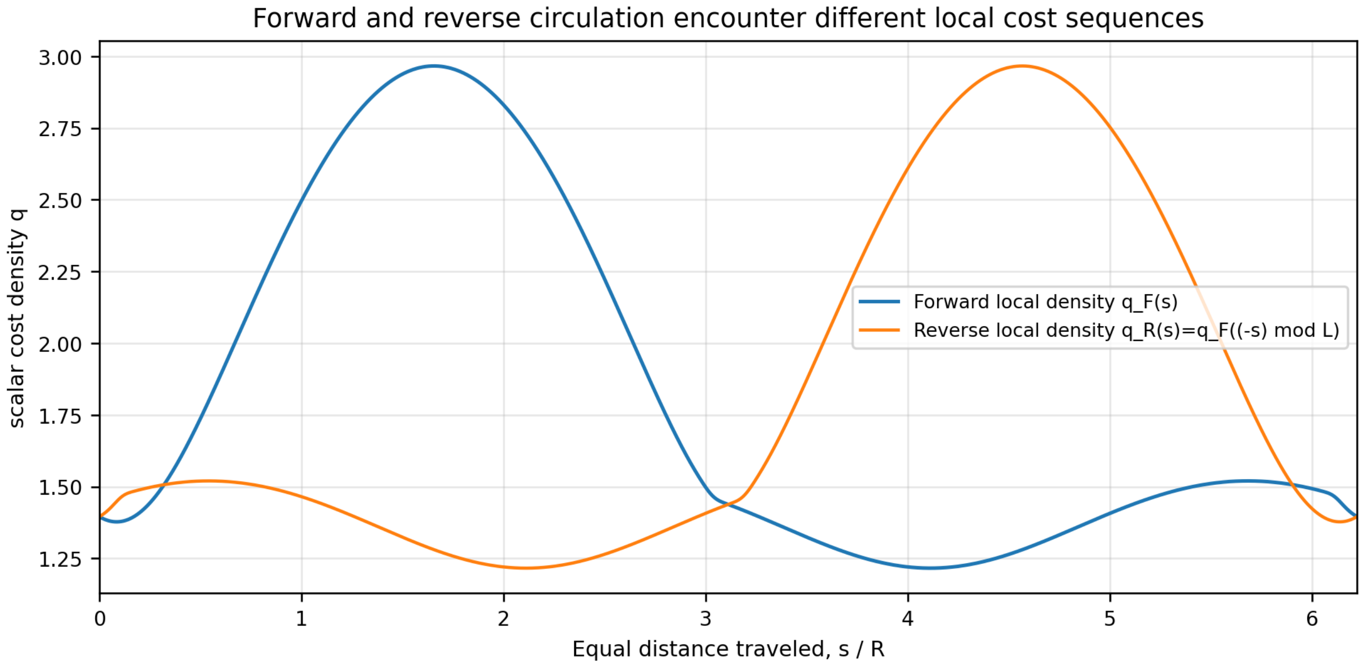

Figure 7. Local scalar density encountered after equal traveled distance. The forward and reverse curves are periodic reversals of one another, not identical sequences. The large difference in ordering is inherited from the high-cost and low-cost meridians.

7. Cumulative Hysteresis Theorem

Examining the forward and reverse circulation reveals a closed cumulative loop. The correct conclusion is not that the completed costs differ. The correct conclusion is that the cumulative scalar cost does not retrace the same path.

Let q(s) be the scalar cost density around the forward-oriented wire coordinate, with total length L. Then the forward and reverse cumulative functions are:

```
C_F(s) = integral_0^s q(u) du
C_R(s) = integral_0^s q(L-u) du = C_loop - C_F(L-s)
```

Because this route is not reflection symmetric about the chosen start point, q(s) is not equal to q(L-s). Therefore C_F(s) is not equal to C_R(s), even though both functions finish at the same completed-loop value:

```
C_F(L) = C_R(L) = 11.276251390281
H(s) = C_F(s) - C_R(s)
H(0) = H(L) = 0
```

After removing the duplicate endpoint before periodic quadrature, the maximum cumulative separation is 2.748220082602 at exactly half the loop, s=L/2=3.111510355488 R. At that point C_F=7.012235736442 and C_R=4.264015653840. The enclosed signed cumulative-cost area is 7.596421328302 cost-distance units, while the absolute enclosed area is 7.617373149318.

Thus the coil exhibits circulation-dependent cumulative hysteresis: forward and reverse propagation encounter different cost histories and converge only after completing the loop. It is not a difference in the final scalar integral. It is a difference in the entire intermediate accumulated state. Any process dependent on propagation time, phase accumulation, thresholds, nonlinear impedance, switching, or state retention can therefore distinguish the two circulation directions despite identical completed-loop cost.

quantity	value
closed-loop cost	11.276251390281409
maximum |C_F-C_R|	2.748220082602275
distance of maximum	3.111510355487929 R

quantity	value
C_F at maximum	7.012235736441801
C_R at maximum	4.264015653839526
maximum / total	24.371752522038 percent
minimum separation	-0.018374745832294 at 0.319794119869593 R
signed loop area	7.596421328301542
absolute loop area	7.617373149317956
zero crossings, s/R	0.000000000000, 0.500799715409, 5.722220995567, 6.223020710564

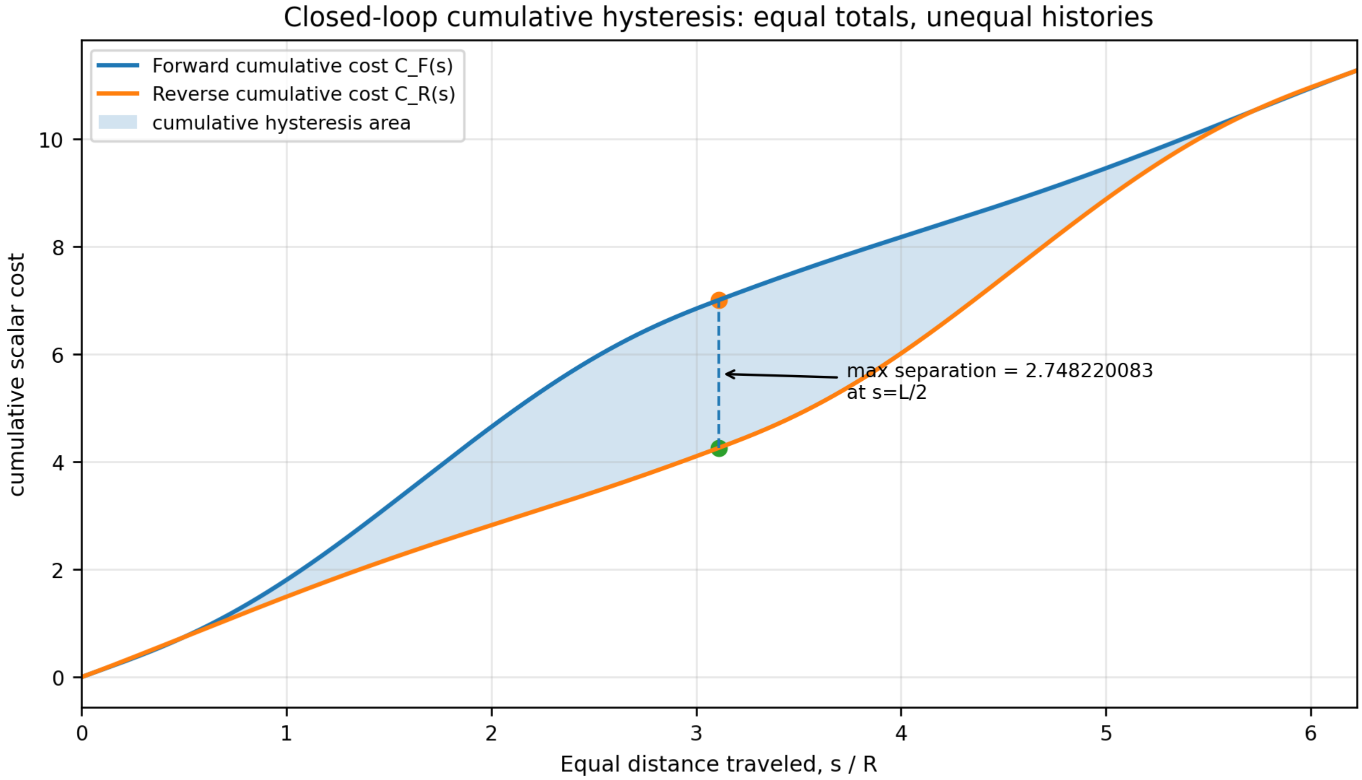

Figure 8. Forward and reverse cumulative costs compared at equal traveled distance. The shaded region is the cumulative hysteresis loop. The largest vertical separation occurs at exactly half the loop.

8. Mathematical Structure of the Hysteresis Loop

The difference function has a useful closed form. Substituting the reverse identity gives H(s)=C_F(s)+C_F(L-s)-C_loop. This immediately proves that H(s)=H(L-s): the cumulative separation is mirror symmetric about half the loop even though the local density itself is not. Differentiation gives H prime(s)=q(s)-q(L-s). Therefore the loop grows wherever the forward circulation is currently encountering a larger local cost than the reverse circulation and shrinks wherever the ordering is reversed.

```
HYSTERESIS IDENTITIES
H(s) = C_F(s) + C_F(L-s) - C_loop
H(L-s) = H(s)
H'(s) = q(s) - q(L-s)
H(L/2) = 2 C_F(L/2) - C_loop
```

The small initial negative lobe, whose minimum is -0.018374745832 at s=0.319794119870 R, records the local bridge and early-meridian ordering near the selected start point. The dominant positive lobe records the much larger high-meridian versus return-meridian imbalance. The signed area is origin dependent because moving the start point cyclically permutes the order in which costs are accumulated. The completed-loop cost is origin independent.

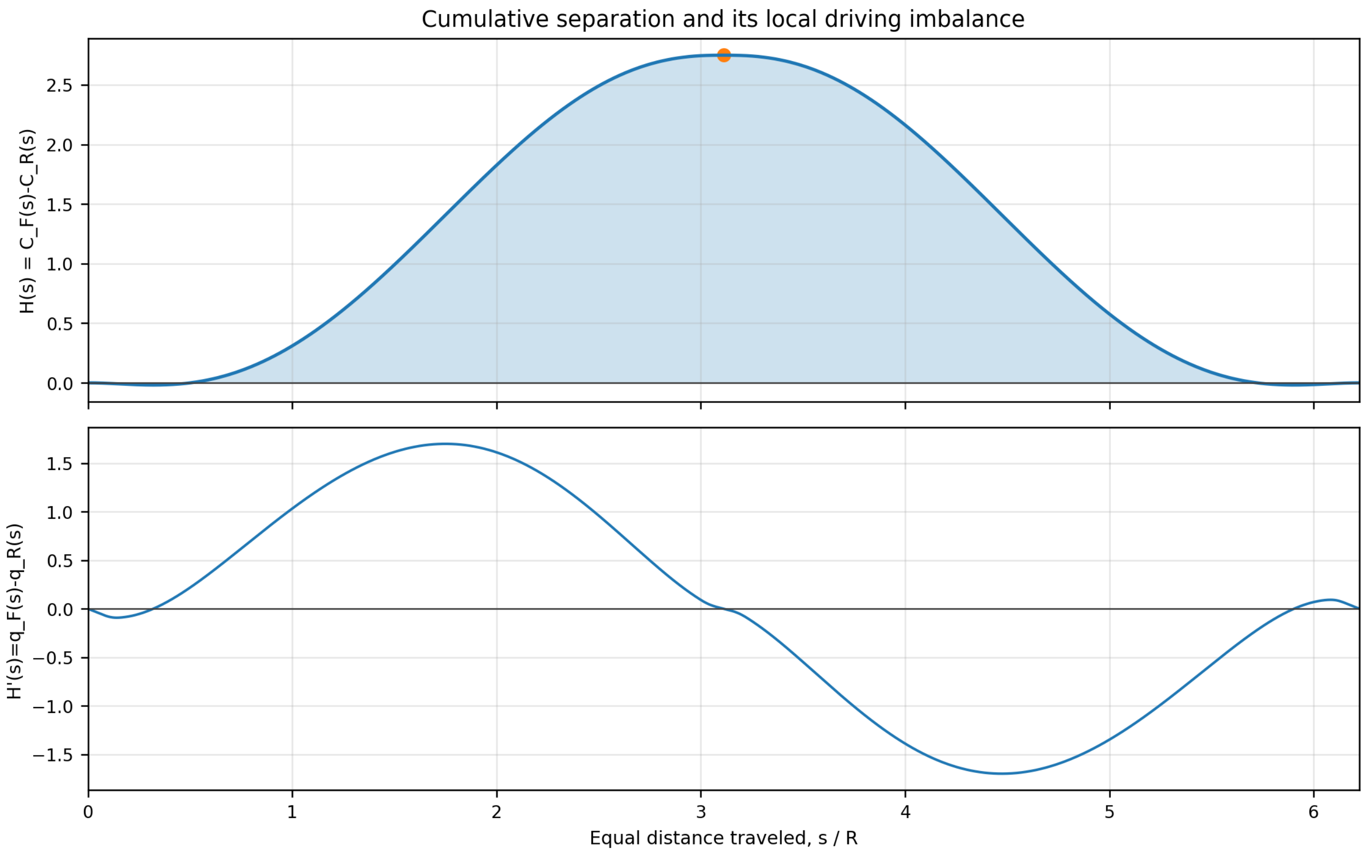

Figure 9. Upper panel: cumulative separation H(s). Lower panel: its derivative, which is exactly the forward-minus-reverse local density imbalance. The maximum at half the loop is stationary because the two circulation directions meet opposite points of the same periodic sequence.

9. Equal-Distance and Same-Location Registration

Two comparisons must be kept distinct. Equal-distance registration compares the two circulations after the same elapsed path length. This is the comparison that produces the closed loop C_F(s) versus C_R(s). Same-location registration compares the cost required to reach the same physical wire point from the common start while approaching from opposite directions. The reverse distance to the forward coordinate s is L-s, so the reverse cumulative cost to that point is C_loop-C_F(s).

```
REGISTRATION IDENTITIES
Equal-distance reverse: C_R(s) = C_loop - C_F(L-s)
Same-location reverse: C_R,to r(s) = C_R(L-s) = C_loop - C_F(s)
Therefore: C_F(s) + C_R,to r(s) = C_loop
```

The same-location curves are complementary rather than retraced. A physical point that is expensive to reach in the forward direction is inexpensive to reach in the reverse direction by exactly the amount required to keep their sum equal to the full-loop cost. This complementarity is the spatial expression of the same ordered accumulation.

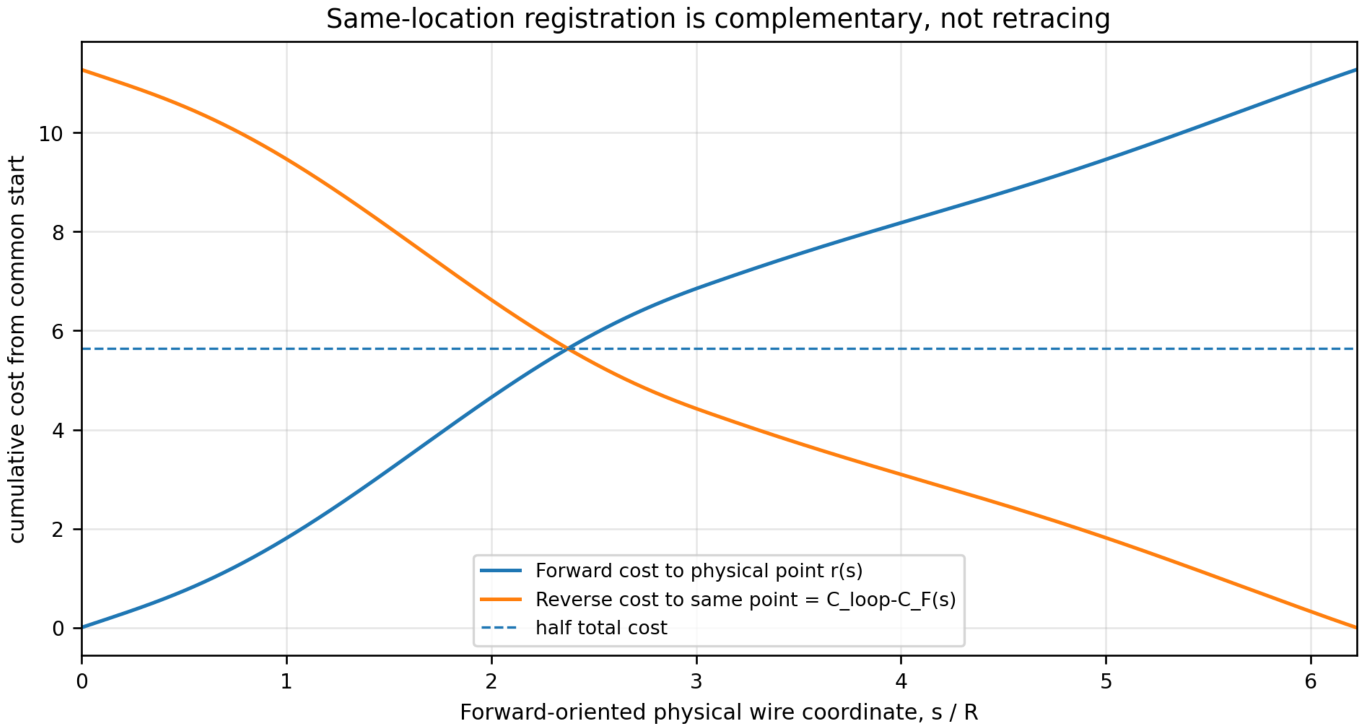

Figure 10. Cumulative costs registered to the same physical wire coordinate. The forward cost and reverse cost from the common start add to the closed-loop total at every point.

10. Physical Interpretation for a Wire Coil

The bare scalar cost q is even under reversal: it assigns one positive value to each spatial direction n regardless of the direction in which the wire is traversed. Therefore the final one-turn scalar integral is reciprocal. The cumulative state is different because integration is ordered. A signal packet, phase front, charge-density perturbation, thermal front, or internal switching variable experiences a different sequence of local costs when circulation reverses.

A linear memoryless series law proportional to q produces the same total resistance in both directions, although the spatial distribution of voltage drop and heating is asymmetric. A stateful law can convert the cumulative hysteresis into a measurable directional response. Examples include a phase threshold reached before or after a nonlinear element, a local constitutive parameter that depends on accumulated phase, a switching element whose state changes after a cumulative-cost threshold, or a relaxation process whose state evolves while the packet traverses the loop.

This distinction is foundational. A direction-odd final integral is not required for direction-dependent intermediate dynamics. The minimal ingredients are: a spatially asymmetric scalar density, a closed path with a specified start or injection point, ordered traversal, and a response law that retains or acts on accumulated state before the loop closes.

```
INTERPRETIVE BOUNDARY
Memoryless terminal law: final response depends only on C_loop -> no directional total
difference
Stateful transport law: response depends on C(s), threshold crossings, or relaxation history
-> directional intermediate difference
Measured hysteresis source: path ordering, not a sign reversal of q
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11. Scaling Laws

For a geometrically similar coil placed on a physical sphere of radius R_phys, every path length scales by R_phys. Since q is dimensionless, cumulative cost also scales by R_phys. The maximum cumulative separation is therefore 2.748220082602 R_phys in cost-length units, and the signed loop area scales as 7.596421328302 R_phys squared.

For N identical series turns concatenated in the same orientation, the total one-way cost scales linearly with N. Within each turn, the same normalized hysteresis profile repeats. If the complete N-turn series path is treated as one cumulative coordinate, both axes of the cumulative plot expand with N and the total geometric area over the concatenated coordinate scales as N squared. Ratios, fractional separation, and the single-turn normalized shape remain unchanged.

quantity	one normalized turn	physical radius R_phys	N identical series turns
length	6.223020710976	6.223020710976 R_phys	6.223020710976 N R_phys
completed cost	11.276251390281	11.276251390281 R_phys	11.276251390281 N R_phys
maximum cumulative separation	2.748220082602	2.748220082602 R_phys	2.748220082602 N R_phys
signed loop area	7.596421328302	7.596421328302 R_phys^2	7.596421328302 N^2 R_phys^2 for concatenated coordinate

12. Experimental and Computational Measurement Pathways

The cumulative effect is most directly tested with a localized excitation and a defined injection point. A pulse is launched in each circulation direction under otherwise identical conditions. The measured observable must be recorded as a function of elapsed path length or time rather than reduced immediately to the completed-loop total.

measurement	direct observable	predicted geometric signature
bidirectional phase pulse	phase versus arrival time around distributed taps	forward and reverse traces separate and reunite after one loop
distributed voltage sensing	cumulative voltage drop from common injection point	same-location traces are complementary; equal-distance traces form the hysteresis loop
threshold element	distance or time at first threshold crossing	direction-dependent crossing caused by unequal cumulative history
relaxing internal state	state variable versus propagation time	different transient trajectory with equal final scalar loop cost
numerical verification	periodic quadrature of q along the wire	C_F(L)=C_R(L), H(L/2)=2.748220082602, signed area=7.596421328302

A completed-loop measurement alone is insufficient because it intentionally discards the intermediate state. The decisive measurement is the trajectory of the accumulated observable.

13. Reproducibility and Numerical Validation

All reported values are generated from the attached route JSON, the smoothed coil timeline, and periodic quadrature. The quadratic metric is symmetric positive definite. Every route sample remains on the unit sphere to numerical precision. The closed centerline and tangent are continuous. The cumulative identities are verified independently from the direct reverse quadrature and from C_loop-C_F(L-s).

validation quantity	result
periodic sample intervals	14400
sample spacing	0.000432154216039990 R
C_F(L)-C_R(L)	7.461e-14
reverse identity max error	1.137e-13
H(s)-H(L-s) max error	1.081e-13
endpoint H(L)	7.461e-14
route radius max deviation	3.331e-16
source route SHA-256	c59612142e6f9aa2925ad4b67a367bb055812e0fb82a7485ab14b24a4d514c51
coil timeline SHA-256	3b02600511a74b344a89449fd15e0a336d43c80da49cf00279b9fa861195a669

14. Foundational Consequences

consequence	meaning
Final reciprocity and trajectory reciprocity are different statements.	A scalar closed-loop integral can be identical under reversal while the accumulated trajectory remains direction dependent.
A scalar field can support geometric cumulative hysteresis.	The hysteresis resides in the ordered integral state, not in a direction-odd local scalar.
The start point becomes an operational boundary condition.	Cyclically moving the injection point changes the signed cumulative-loop area while preserving the completed-loop total.
Stateful elements convert geometry into directional response.	Thresholds, memory, phase-sensitive coupling, and relaxation can act before the cumulative traces reunite.
Intermediate-state measurements are primary observables.	Reducing the experiment to one final loop number erases the phenomenon being tested.

The resulting paradigm is a transport geometry in which the path is not characterized only by its total integral. It is characterized by the ordered cumulative state over the full circulation. That state contains information that is invisible to the endpoint total and can be coupled to physical dynamics.

Appendix A. Complete Cell3 Half-Step Node Registry

Every node is generated exactly by x=0.5(i a+j b+k c), with i,j,k in {-1,0,1}. Coordinates are native ICRS Cell3 units. These coordinates are not route-sphere coordinates.

i	j	k	lambda	ICRS x native	ICRS y native	ICRS z native
-1	-1	-1	-3	-0.01097490249996122	0.00738598312742294	0.03966191101193925
-1	-1	0	-2	-0.01129009224263490	-0.00474287788114100	0.01819973372639234
-1	-1	1	-1	-0.01160528198530859	-0.01687173888970493	-0.00326244355915457
-1	0	-1	-2	-0.00987552951579954	0.01383706825676921	0.02262866024391139
-1	0	0	-1	-0.01019071925847322	0.00170820724820527	0.00116648295836448
-1	0	1	0	-0.01050590900114691	-0.01042065376035866	-0.02029569432718243
-1	1	-1	-1	-0.00877615653163786	0.02028815338611548	0.00559540947588353
-1	1	0	0	-0.00909134627431155	0.00815929237755154	-0.01586676780966337
-1	1	1	1	-0.00940653601698523	-0.00396956863101239	-0.03732894509521029
0	-1	-1	-2	-0.00078418324148799	0.00567777587921766	0.03849542805357477
0	-1	0	-1	-0.00109937298416168	-0.00645108512934627	0.01703325076802786
0	-1	1	0	-0.00141456272683536	-0.01857994613791020	-0.00442892651751905
0	0	-1	-1	0.00031518974267369	0.01212886100856393	0.02146217728554691
0	0	0	0	0.00000000000000000	0.00000000000000000	0.00000000000000000
0	0	1	1	-0.00031518974267369	-0.01212886100856393	-0.02146217728554691
0	1	-1	0	0.00141456272683536	0.01857994613791020	0.00442892651751905
0	1	0	1	0.00109937298416168	0.00645108512934627	-0.01703325076802786
0	1	1	2	0.00078418324148799	-0.00567777587921766	-0.03849542805357477
1	-1	-1	-1	0.00940653601698523	0.00396956863101239	0.03732894509521029
1	-1	0	0	0.00909134627431155	-0.00815929237755154	0.01586676780966337
1	-1	1	1	0.00877615653163786	-0.02028815338611548	-0.00559540947588353
1	0	-1	0	0.01050590900114691	0.01042065376035866	0.02029569432718243
1	0	0	1	0.01019071925847322	-0.00170820724820527	-0.00116648295836448
1	0	1	2	0.00987552951579954	-0.01383706825676921	-0.02262866024391139
1	1	-1	1	0.01160528198530859	0.01687173888970493	0.00326244355915457
1	1	0	2	0.01129009224263490	0.00474287788114100	-0.01819973372639234
1	1	1	3	0.01097490249996122	-0.00738598312742294	-0.03966191101193925

Appendix B. Exact Route and Lune Parameters

field	value
Earth CMB unit vector	[-0.9707708558359351, 0.20734800911098614, -0.12087492948199192]
apex RA, Dec	167.943299879827578, -6.942599922129508 deg
antapex RA, Dec	347.943299879827578, 6.942599922129508 deg
high-meridian b_E	[-0.2007427663336889, -0.4254133415787006, 0.8824544354081642]
high-meridian plane normal	[0.1315533626490182, 0.8809258152220052, 0.4546024866311025]
return-meridian b_R	[0.13236005555333083, 0.8826304811640974, 0.451047945803939]
return-meridian plane normal	[0.2002117907454136, 0.4218651879898889, -0.8842765415913567]
projective plane angle	89.769450000753878 deg
closed-lune solid angle	3.133544940435163 sr
sphere fraction	0.249359583335427
unsmoothed cost difference	2.742391870247186

Appendix C. Machine-Readable Companion

The companion file K-Field_hysteresis_closed_CMB_route_1784658249.json contains the authoritative source snapshot, phase metric eigensystem, complete Cell3 vertices and 27-node registry, six face support planes, route construction, segment statistics, 14,401-point closed cumulative timeline, equal-distance forward and reverse traces, same-location complementary trace, all hysteresis metrics, dimensional scaling rules, source hashes, and figure hashes.

JSON section	contents
phase_metric	G_phase matrix, eigenvalues, eigenvectors, determinant, trace, condition number
cell3_native_ICRS	basis, lengths, face areas, normals, support planes, vertices, all 27 nodes
closed_route	CMB endpoints, meridian planes, bridges, segment statistics
cumulative_hysteresis.metrics	closed total, maximum separation, loop areas, crossings, numerical identity errors
cumulative_hysteresis.timeline	distance, segment, position, tangent, q_F, q_R, C_F, C_R, H, same-location reverse cost for every sample
figure_manifest	file name and SHA-256 for each data-generated figure

Archival Colophon

Field	Entry
Document title	K-FIELD: CUMULATIVE HYSTERESIS ON THE CLOSED EARTH-CMB ROUTE
Project title	K-Field cumulative transport geometry in the authoritative Earth-CMB orientation
Principal Investigator	P. Dwayne Esterline
Credentials	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx
Output PDF	K-Field_hysteresis_closed_CMB_route_1784658249.pdf
Machine-readable companion	K-Field_hysteresis_closed_CMB_route_1784658249.json
Generation epoch	1784658249
Page size	US Letter
Coordinate frame	ICRS Cartesian; route sphere normalized to R=1; Cell3 retained in separate native units
Fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Source route	K-Field_max_residual_closed_CMB_route_1784650136(2).json
Source route SHA-256	c59612142e6f9aa2925ad4b67a367bb055812e0fb82a7485ab14b24a4d514c51